lotor lab

Innovating on R&D through
Adaptable Evidence Generation.

Our DNA

Our View

Our Proof

Noun. The brightest star in the constellation Canis Minor. Constellations transform isolated points of light into recognizable structure, giving individual stars meaning.

Procyon lotor

Noun. The scientific name for the common North American racoon, defined by adaptability, thriving in complex, unpredictable environments through intelligence, observation, and precise problem-solving. It succeeds where conditions are unpredictable and paths are undefined.

Inspired by the clever *Procyon lotor*, we built ***lotor lab*** – serving our clients with the same resiliency and adaptability.

We excel in supporting first-in-class therapies where **no clear precedent exists**—applying resourceful, original thinking to identify the right clinical and real-world evidence that **maximizes probability of success**.

Regulatory approval is rarely driven by a single point, but rather by a carefully assembled **constellation of evidence**.

We are

lotor lab

MICHELLE GUO

Co-Founder & CEO

Previously Clinical R&D Portfolio Leader at Deloitte

Computational Biology, UChicago | Epidemiology, Yale

ROHAN BHUTKAR

Co-Founder & CTO

Previously Lead Engineer in Clinical R&D at Deloitte

Interdepartmental Immunology, McGill | Epidemiology, Imperial

Our DNA

Our View

Our Proof

$2.23B

to bring one drug asset
from discovery to launch

Drug development is no longer a pipeline. It’s a high-stakes bet.

Over the past decade, the cost per approved drug drug has increase 7% while IRR for the top 20 biopharma companies has declined to 5.9%

Failure is expensive.
A cohesive evidence strategy needs to exist from day one.

The Consequence:

To ensure success, you cannot rely on precedent, as it <u>does not exist</u> for novel therapies

Therapies are changing, leading to more complex trials with a higher bar for evidence

Increased pressure for novel therapies including personalized medicine and cell & gene therapy

R&D costs are increasing

lotor lab

- We work with our clients to write the playbook they need for success.
  We don’t follow existing playbooks
- Creating evidence strategies for novel therapies that don’t fit traditional regulatory frameworks
- Define regulatory pathways and design evidence packages to align clinical strategy with commercial success

Our constellation of **capabilities** support this multi-faceted approach to intelligent evidence generation and regulatory success

Architecting the Foundation

Building core clinical and regulatory strategy including protocol optimization, from endpoint selection to SoA design

Translate for Access

Bridge evidence gaps with RWE and value demonstration aligning with payer priorities

Drive Lifecycle Value

Ensuring long-term success through ongoing evidence generation and stakeholder engagement

Enabling Advanced AI/ML Capabilities

Experience building and scaling enterprise-grade agentic AI systems so support evidence generation including eAuthoring, public data scraping, horizon scanning and internal and external multidata source networking

Our DNA
Our View
Our Proof

Key Quals

- Supporting Executive Leadership
- Protocol Development
- Pre-Launch Strategy
- Clinical Program Design
- Meta-analysis and Data Exchange
- Post-Launch Execution
- Economic Modeling
- Regulatory Strategy
- Payer & HTA Strategy

De Novo regulatory strategy for a first-in-class PDT

Situation

- First in class Prescription Digital Therapeutic (PDT) required a credible FDA package despite limited precedent
- Evidence was distributed across multiple studies and needed to be packaged into a cohesive submission strategy to achieve a broad label

Approach

- Evaluated regulatory pathway options and organized De Novo pathway support
- Designed evidence package and submission narrative for FDA engagement
- Synthesized findings across three clinical trials
- Mined precedent cases to strengthen classification rationale
- Developed investor materials aligned to the regulatory story

Outcome

Established a clear regulatory pathway for a novel product category

Improved coherence and defensibility of the submission narrative

Enabled consistent communication across regulatory and investor audiences

Translating clinical trial evidence into a payer-relevant value narrative

Situation

- Client had evidence from three trials but lacked a unified story across clinical, statistical, and payer audiences
- Existing trial endpoints did not directly connect to reimbursement and economic value measures

Approach

- Reviewed and organized data room materials across three studies
- Built a clinical and biostatistical narrative based on evidence strength and relevance
- Supported HCP engagement and evidence validation
- Created charts, diagrams, and modality visuals for cross-stakeholder communication
- Built a crosswalk from trial endpoints to EQ-5D, then to QALYs and ICER for payer relevant claims

Outcome

Unified fragmented evidence into a coherent stakeholder-ready narrative

Created a novel bridge from clinical outcomes to payer value measures

Strengthened support for reimbursement and cost-effectiveness discussions

Pre-Launch Medical Affairs and RWE Strategy for Rare Disease Therapy

Situation

- Small biotech needed to build a compelling evidence narrative to support payer formulary positioning and HCP uptake ahead of launch
- Required a clearer view of disease burden and patient need in a rare disease population
- Also needed targeted KOL engagement to inform medical affairs strategy and market readiness

Approach

- Used RWE claims data to assess burden of disease and sharpen the patient narrative
- Built RWE evidence inputs for the PIE deck and broader medical affairs story
- Conducted KOL targeting to prioritize key stakeholders and strengthen pre-launch engagement strategy
- Positioned the evidence narrative to support both payer value communication and HCP education

Outcome

- **Strengthened pre-launch evidence narrative around disease burden, unmet need, and product value**
- **Supported more targeted medical affairs and stakeholder engagement planning**
- **Laid the foundation for payer access discussions and early HCP uptake**

Post-Launch Real-World Evidence and Outcomes Assessment for Rare Disease Therapy

Situation

- Following launch, uncovering how the therapy was performing in the real world relative to trial expectations and competitor products
- Needed evidence to support ongoing payer positioning and quantify the therapy's real-world clinical and economic value
- Identified downstream drivers of comorbidities, mortality, and cost burden in the target population

Approach

- Analyzed specialty pharmacy data to assess real-world use of the therapy, including titration, adverse events, and discontinuation patterns
- Compare post-launch utilization / outcomes against trial data and competitors
- Conducted additional claims data analysis using machine learning methods to identify drivers of comorbidities, mortality, and healthcare costs
- Used findings to support understanding of the potential cost offsets associated with the client's therapy

Outcome

Generated a clearer view of real-world therapy performance and competitive positioning post-launch

Identified key clinical and economic drivers within the patient population

Built evidence to support future payer value demonstration and broader post-launch medical affairs strategy

Asset Assessment and Clinical Strategy Diligence for BD Decision-Making

Situation

- Client needed ongoing assessment of external assets for acquisition or partnership potential
- Required a structured view of each asset's scientific, clinical, strategic, and commercial viability to inform executive go / no-go decisions
- Needed to understand both the quality of the current development approach and the asset's future value potential

Approach

- Conducted end-to-end asset assessments spanning preclinical data, clinical evidence, and development strategy to identify strengths, risks, and gaps
- Developed net-new clinical and evidence-generation strategies to assess future development pathways and value creation opportunities
- Built financial models to size the opportunity, including patient funnel, addressable population, uptake, and ROI potential
- Synthesized scientific, clinical, and commercial findings into an executive-level recommendation on asset attractiveness

Outcome

Enabled leadership to make more informed acquisition and partnering decisions

Identified key risks and value-creation levers across asset development and evidence strategy

Provided an integrated view of clinical potential and commercial upside to support go / no-go investment decisions

Evidence Lead for Clinical-Stage Combination Therapy

Situation

- Client needed an evidence lead to support a clinical-stage combination therapy through key early development milestones
- Program required an integrated evidence strategy spanning Q-Sub development, protocol authoring, and overall clinical evidence planning
- The asset was part of a partnership, requiring alignment across both internal and partner organizations

Approach

- Served as evidence lead for the program, owning development of the Q-Sub, protocol, and clinical evidence strategy
- Balanced competing priorities to manage cross-functional inputs across manufacturing, medical, safety/PV, biostats, and other stakeholders
- Design a SoA that minimized patient burden & cost while preserving rigor
- Coordinated with the partner organization to incorporate external inputs and align on evidence requirements for the combo therapy

Outcome

Advanced a cohesive, cross-functional evidence strategy for a complex clinical-stage asset

Enabled alignment across internal teams and partner stakeholders on protocol and evidence priorities

Supported a more efficient study design that balanced operational feasibility, patient experience, and scientific robustness

Pricing and Market Access Strategy for New Immunology Therapy

Situation

- Developed pricing strategy for a therapy entering a crowded market
- Required a clear view of how the evidence package, expected clinical outcomes, and HTA assessments would shape pricing flexibility and access
- Pricing complexity was heightened by multiple doses, titration patterns, and indications across multiple therapeutic areas

Approach

- Assessed clinical and HTA outcomes for price and access implications
- Built a revenue model to define price thresholds by market, balancing broad coverage and minimal restrictions with WAC
- Estimated required rebate levels to support target access outcomes
- Developed a competitive landscape of comparator products, including utilization, market share, and cost dynamics, to model expected share capture
- Designed a complex utilization-based rebate approach to account for varying doses, titration patterns, and cross-indication pricing dynamics

Outcome

- **Defined a market-specific pricing strategy to optimize access while preserving value**
- **Quantified rebate expectations and pricing thresholds needed to support payer coverage goals**
- **Built a more realistic view of competitive share capture in a crowded market**
- **Supported decision-making for a complex, multi-indication pricing and contracting strategy**

Global Clinical Program Design for First-in-Human Gene Therapy

Situation

- Clinical development strategy that could safely and credibly reach the target pediatric population despite regulatory constraints
- FDA initially did not permit a pediatric trial, requiring an alternative pathway to progress development

Approach

- Designed E2E first-in-human program with age-descending strategy
- Established the initial development pathway through Health Canada with enrollment descending from 24+ to 18–24 followed by MHRA acceptance of further age descent from 18 to 15
- Structured the overall program to enable progressive expansion into the intended pediatric population while maintaining clinical rigor and regulatory credibility

Outcome

Created a viable global clinical development pathway for a first-in-class gene therapy

Enabled staged progression from adults to adolescents as part of a broader age-descending strategy

Positioned the program to reach the target pediatric population through a structured, regulator-informed development plan

Regulatory and Evidence Strategy for GTx Pediatric Expansion

Situation

- Built an evidence base to expand from adolescent into pediatric
- Regulators required a compelling justification for pediatric inclusion and for the use of a non-traditional study design in a rare disease
- The program needed additional evidence beyond trial data to support regulatory decision-making

Approach

- Engaged FDA in parallel during trial execution to support transition from adolescents to pediatric patients
- Committed to a natural history study to support pediatric expansion discussions with FDA
- Used burden-of-disease evidence to persuade health authorities that a delay-to-treat arm was sufficient in place of a traditional control arm

Outcome

- **Strengthened the regulatory case for pediatric expansion using a broader evidence package**
- **Secured health authority acceptance of a more feasible study design in a rare disease setting**
- **Contributed to a successful approval by combining trial evidence with RWE and natural history support**